

Aram Ansary Ogholbake

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Research Profile

Machine learning researcher with interdisciplinary background in computer science, mathematics, and finance. Experienced in developing machine learning and deep learning methods for complex, high-dimensional data, including transcriptomic, proteomic, medical imaging, and multimodal clinical data, as well as textual and structured patent data.

Education

Ph.D. in Computer Science, University of Kentucky August 2026

Master of Science in Financial Mathematics, University of Kentucky May 2020

Bachelor of Science in Mathematics, University of Kentucky May 2018

Research & Teaching Experience

Graduate Research/Teaching Assistant, University of Kentucky Aug 2020 – June 2026

Department of Computer Science and Institute for Biomedical Informatics

- Developed neural network and patent-language-model approaches for patent approval-time prediction and semantic novelty characterization, integrating LLM-based text representations with structured patent data.
- Developed a multimodal deep learning framework combining CT imaging and structured clinical data for edema classification.
- Developed interpretable models that reduce peripheral-blood gene-expression profiles to two-gene symbolic diagnostic formulas, with evaluation on independent bulk and single-cell RNA datasets across multiple cancer types.
- Developed unsupervised deep learning models for circadian phase prediction from high-dimensional transcriptomic and proteomic data, including the PENN and PROTECT frameworks, and evaluated them across independent human, mouse, plant, and other organism datasets.
- Developed interpretable compact gene-expression classifiers for cancer detection, including a five-gene hepatocellular carcinoma model.
- Conducted computer vision research in image segmentation, change detection, regression from weak labels, and single-image depth estimation.
- Contributed to research on fine-grained property-value assessment using probabilistic disaggregation and remote-sensing imagery.
- Served as a graduate teaching assistant in the Department of Computer Science.

Graduate Teaching Assistant, University of Kentucky Spring 2020

Department of Physics & Astronomy

Key Technical Skills

Programming & tools Python, C++, C, R

ML Tools PyTorch, PyTorch Lightning, TensorFlow, Keras, scikit-learn, Weights & Biases, NumPy, pandas

AI Methods Transformers, large language models, multimodal learning, attention mechanisms, self-supervised learning, unsupervised learning

Bioinformatics Bioconductor, PyDESeq2, limma, Scanpy, GSEAPy

Visualization Matplotlib, Seaborn, ggplot2, ggrepel

Computing Linux, Git/GitHub, Conda, HPC/GPU computing

Selected Publications

Journals

- **Ansary Ogholbake, A.,** & Cheng, Q. (2025). Five-Gene Expression Formula Accurately Detects Hepatocellular Carcinoma Tumors. *Biotechnology Journal*, 20(7), e70073.
- **Ansary Ogholbake, A.,** & Cheng, Q. (2025). PROTECT: Protein Circadian Time Prediction Using Unsupervised Learning. *iScience*, 28(10).
- **Ansary Ogholbake, A.,** & Cheng, Q. (2024). Gene Expression Clock: An Unsupervised Deep Learning Approach for Predicting Circadian Rhythmicity from Whole Genome Expression. *Neural Computing and Applications*, 36(33), 20653–20670.
- **Ansary Ogholbake, A.,** & Cheng, Q. (2023). PENN: Phase Estimation Neural Network on Gene Expression Data. In *The International Conference on Deep Learning, Big Data and Blockchain* (pp. 59–67).

Conference

- Archbold, C., Brodie, B., **Ansary Ogholbake, A.,** & Jacobs, N. (2023). Fine-Grained Property Value Assessment Using Probabilistic Disaggregation. In *IGARSS 2023 – IEEE International Geoscience and Remote Sensing Symposium* (pp. 5312–5315). IEEE.

Preprint

- **Ansary Ogholbake, A.,** Choi, H., Brandenburg, S., Antuna, A., Al-Sharshahi, Z., Cox, M., Ahmed, H., Frank, J., Millson, N., Bauerle, L., et al. (2026). A Multimodal Deep Learning Framework for Edema Classification Using HCT and Clinical Data. *arXiv preprint*.

Patent

Cancer Classification Using Peripheral Blood Gene Expression

U.S. Patent Application No. 19/717,266, filed June 23, 2026. (*Patent Pending*)

Honors & Awards

- **Outstanding Student Paper Award**, Department of Computer Science, University of Kentucky, 2025.
- **Master of Science in Finance Scholarship**, University of Kentucky.
- **Dean's List**, University of Kentucky.